

8D PROBLEM SOLVING REPORT

Problem Title:	Elevated Iron Content in H-Acid Batches B0351–B0430
Product:	H-Acid (1-amino-8-naphthol-3,6-disulfonic acid)
Report No.:	8D-2023-001
Date Opened:	2023-01-15
Date Closed:	2023-05-20
Prepared By:	Supply Quality Engineer — Techno IT Hub
Status:	CLOSED

D1: Define the Team

Team Leader: Supply Quality Engineer (SQE)

Members: Production Manager, Maintenance Engineer, QA Lab Analyst, Procurement (Supplier D liaison)

Sponsor: Plant Quality Manager

Meeting cadence: Weekly during investigation; bi-weekly for closure verification

D2: Describe the Problem

What: Iron content (Iron_ppm) in H-Acid batches exceeded the USL of 10 ppm starting at Batch B0351 (2022-12-17), with peak values reaching 16+ ppm. A total of 80 batches (B0351–B0430) were affected.

Who Detected: SPC monitoring via I-MR control chart in Minitab flagged the first Test 1 violation (1 point $>3\sigma$) at Batch B0355.

When: Drift began ~Batch #351 (Dec 2022); confirmed OOC pattern by Jan 2023; corrected by Batch #431 (Mar 2023).

Impact: Ppk for Iron_ppm dropped from 1.18 (Batches 1–350) to 0.29 (all 500 batches). 8 distinct OOC signal events logged. Customer complaint risk elevated.

Magnitude: ~16% of total production volume (80/500 batches) potentially off-spec for iron content.

D3: Interim Containment Actions

1. Segregated and quarantined all batches B0351–B0430 pending iron content re-test (100% inspection by QA lab).
2. Issued customer notification for any shipments from affected period; initiated hold on pending shipments.
3. Increased sampling frequency to every-batch iron ppm testing (from every 5th batch) during investigation.

D3: Interim Containment Actions

4. Issued supplier advisory to Supplier D (Iron Scrap) requesting CoA for all recent lots.

Effectiveness confirmed: Iron testing frequency increased within 3 days of problem identification.

D4: Define & Verify Root Cause

Root Cause (Confirmed): Corrosion of the cast-iron reactor lining in Reduction Reactor R-02. Progressive lining wear between scheduled maintenance windows (6-month interval) allowed iron particles to leach into the batch, particularly under the acidic conditions of the iron/acid reduction step.

Contributing Cause: Supplier D (Iron Scrap) lot variability — incoming purity 91.0% vs. internal spec $\geq 93\%$. Higher impurity iron scrap accelerated lining corrosion chemistry.

Fishbone Analysis: Machine branch — reactor lining wear; Material branch — Supplier D iron scrap quality; Method branch — maintenance interval too long; Measurement branch — no in-process iron sensor (only batch-end QA lab test).

Verification: Reactor R-02 lining inspection (Jan 2023) confirmed $>40\%$ wall-thickness reduction at bottom wear zone. Iron ppm decreased to baseline (mean 7.3 ppm) immediately after lining replacement (Batch B0431 onward).

D5: Choose Permanent Corrective Actions

CA-1 (Machine): Replace reactor R-02 cast-iron lining with SS316L stainless steel. Scheduled maintenance window Feb 2023. COMPLETED.

CA-2 (Material / Supplier): Issued Supplier D corrective action requiring (a) incoming purity $\geq 94\%$; (b) lot-level CoA with ICP iron analysis; (c) quarterly supplier audit. Q4-2023 re-audit: Supplier D purity improved to 95.2%, OTD improved to 89%.

CA-3 (Method): Reduce reactor lining inspection interval from 6 months to 3 months (bi-monthly). Updated maintenance PM schedule.

CA-4 (Measurement): Added iron_ppm alert threshold at 8.5 ppm in Minitab SPC control chart (1 sigma warning before USL breach).

D6: Implement & Validate Corrective Actions

Reactor lining replaced Feb 12, 2023 — confirmed by maintenance work order MWO-2023-0212.

Supplier D corrective action plan accepted Mar 1, 2023. Re-audit completed Nov 2023 (Score: 83/100, Grade B).

Maintenance PM schedule updated in CMMS — first new 3-month inspection completed Apr 2023 (no wear noted).

SPC alert threshold updated in Minitab project H-Acid_Quality_Project.mpx.

D6: Implement & Validate Corrective Actions

Validation: Iron_ppm for batches B0431–B0500 (70 post-correction batches): Mean = 7.34 ppm, StdDev = 1.41. All within control limits. Ppk (post-correction only) = 1.16. SUCCESS.

D7: Prevent Recurrence

PFMEA updated: Reduction step iron-contamination row revised — OCC reduced from 7 to 2 (post-CA); RPN reduced from 252 to 72 (GREEN).

Control Plan updated to require every-batch Iron_ppm testing at QA lab.

Supplier Quality Procedure SQP-004 updated: iron scrap suppliers now require ICP-verified CoA per lot.

Lesson learned added to site Quality Management System (QMS): 'Reactor lining inspection frequency must be calibrated against corrosive process chemistry, not fixed time intervals.'

Training: Maintenance and production operators re-trained on reactor lining wear indicators (Mar 2023).

D8: Congratulate the Team & Close

This 8D was formally closed on 2023-05-20 after 70 consecutive post-correction batches confirmed Iron_ppm within control limits (Ppk = 1.16).

Team recognition: Plant Quality Manager issued commendation to the cross-functional team for identifying and eliminating the root cause within the investigation window.

Customer notification: Formal closure letter sent to key customers affected by quarantine hold. Zero customer rejections from this event.

Project artifact: Full SPC analysis, capability before/after comparison, and supplier audit evidence documented in the H-Acid Manufacturing SPC Portfolio (GitHub: h-acid-quality-project).

This 8D report was prepared using synthetic/simulated data for portfolio and interview demonstration purposes. No proprietary employer data is contained herein.